Question Paper

Exam Date & Time: 12-Nov-2019 (10:00 AM - 01:00 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

I Semester of B. Sc. Examination, November, 2019
Course code & Name BS 108 & Biochemistry
Date: 12/11/2019 Day: Tuesday

BIOCHEMISTRY [BS108]

Marks: 70 | **Duration: 180 mins.**

MCQ

Answer all the questions. Section Duration: 30 mins

Choose the correct answer for the following questions.

1) Identify the purine base of nucleic acids in the following (1)
Adenine | Thymine | Cytosine | Uracil

2) Enantiomers are (1)
not chiral. | compounds with very different chemical composition. | molecules with both hydrophobic and hydrophilic characteristics. | a pair of compounds that are non superimposible mirror-images of one another.

3) Group of adjacent nucleotides are joined by (1)
Covalent bond. | Peptide bond. | Ionic bond. | Phosphodiester bond.

4) Nucleotides have phosphates attached at the ___ carbon atom of the sugar. (1)
8' | 5' | 3' | 1'

5) Water molecules are attracted to each other because of the opposite charges created by partial charge separations within the molecules. These attractions are called (1)
Covalent bonds. | Ionic bonds. | Hydrogen bonds | Atomic bonds.

6) The naturally occurring form of amino acid in proteins (1)
L-amino acids only. | D-amino acids only. | Both L & D amino acids. | None of these.

7) Raffinose is a (1)
Monosaccharide. | Oligosacchaaride. | Disacchaaride. | Polysacchaaride.

8) Amino acids are transported to the ribosome for use in building the polypeptide by (1)
mRNA molecules. | tRNA molecules. | rRNA molecules. | DNA polymerase molecules.

9) DNA consists of two antiparallel strands of nucleotide chains held together by (1)
peptide bonds. | hydrogen bonds | polar covalent bonds. | complementary base pairing bonds.

10) A common function of Thiamin, Riboflavin and Niacin is that: (1)
They all are used in synthesis of blood clotting proteins. | They all work as a part of a coenzyme used in energy metabolism. | They all help to strengthen blood vessel walls. | They are used to stabilize cell membranes.

11) The polypeptide-making organelles residing in the cytoplasm are large protein aggregates to which RNA is associated. They are called (1)
Ribosomes. | Golgi bodies. | The endoplasmic reticulum. | Mitochondria.

12) Which one of these vitamins is involved in blood clotting? (1)
Vitamin B_6. | Vitamin B_{12}. | Vitamin E. | Vitamin K.

13) Which one of these vitamins is involved in calcium homeostasis? (1)
Vitamin A. | Vitamin D. | Vitamin K. | Vitamin E.

14) A common lipid for energy storage is (1)
Phospholipid. | Steroid. | Triglycerides. | Cholesterol.

15) The lipid layer that forms the foundation of cell membranes is primarily composed of molecules called ____________. (1)

Phospholipids. Carbohydrates. Fats. Proteins.

16) Which of the following is also known as invert sugar? (1)

Sucrose Dextrose Fructose. Glucose.

17) Which of the following is not a macromolecule? (1)

Water. Proteins. Carbohydrates. Lipids.

18) Name the major storage form of carbohydrates in animals? (1)

Chitin. Cellulose Glycogen. Starch.

19) Glycogen in animals are stored in (1)

Liver and stomach. Liver and bile. Liver and muscles. Liver and adipose tissue.

20) To possess optical activity, a compound must be (1)

A carbohydrate. A hexose. D-glucose. Asymmetric.

I

Answer 5 out of 8 questions.

Instructions:

Make suitable assumptions and draw neat figures wherever required.
Use of non-programmable calculator is allowed.Show necessary calculations.

Answer the following questions as directed (Any 5).

1) How biomolecules can be classified based on their interactions with water? Describe briefly with example. (4)

2) Major component in the body of polar bear is fat. Polar bear fast during hibernation in months of summer, due to scarcity of food. (4)
a) Based on the nature of habitat, that is sea ice; mention and justify the kind of fat that would be predominantly present in the body of an active (swimming or hunting) polar bear.
b) Explain in terms of energy production, why fat is preferred over other biomolecules in polar bear.

3) a) What are hexoses? What are some examples of hexoses with important biological functions? (4)

b) What are pentoses? What are the roles of pentoses in DNA and RNA molecules?

4) What are D and L-isomers? Explain briefly. (4)

5) What are antioxidants? Which vitamins can act as antioxidants? What are the signs and symptoms of vitamin E deficiency? (4)

6) How can condensation reaction result in the formation of macromolecules? Identify the monomers and the kinds of bonds that are formed in the above mentioned process to synthesize gene, glycogen and insulin. (4)

7) Write short notes on Hydrogen bond directionality and strength. (4)

8) What are the different qualitative methods to identify carbohydrates? Write down the name of different quantitative estimation methods for proteins, amino acids, nucleic acids and carbohydrates respectively. (4)

II

Answer 10 out of 14 questions.

Answer the following questions as directed (Any 10).

1) Describe briefly the ether link lipids. (3)

2) What are the medical and biochemical importance of water? (3)

3) What are essential fatty acids? Give names of 3 essential fatty acids? What are the characteristics of sterol? (3)

4) What is the role of vitamin C in the body? What damage is caused by a lack of vitamin C? Why was this deficiency also known as "sailors' disease"? (3)

5) Discuss briefly on noncovalent Interactions and macromolecular structure. (3)

6) What are the differences between DNA and RNA? (3)

7) What is the rule for the pairing of nitrogenous bases within the DNA molecule? What about in RNA molecules? Is this last question relevant? (3)

8) Write a short note on Artificial Sweeteners. (3)

9) What is the relationship between vitamin D and sunlight? What disease is caused by vitamin D deficiency? Which tissue does it affect? (3)

10) What vitamins make up the B complex? What problems are caused by a lack of these vitamins? (3)

11) What are the three main types of RNA and what are their functions? (3)

12) What is the complementary sequence of nitrogenous bases for an AGCCGTTAAC fragment of a DNA chain?
Is there any situation in which DNA is made based on an RNA template?
Which is the enzyme involved? (3)

13) Why do membranes consist of lipid bilayers rather than lipid monolayers? (3)

14) Write short notes on Nucleosome model of chromosome. (3)

-----End-----